

# Motor Control Board

Variant: DRAFT

2026-08-27

Rev + (Unreleased)

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## TOP VIEW

kibot\_Mimage\_png\_3d\_viewer\_angled\_top

## BOTTOM VIEW

kibot\_Mimage\_png\_3d\_viewer\_angled\_bottom

## DESIGN CONSIDERATIONS

DESIGN NOTE:  
Example text for  
informational design  
notes.

DESIGN NOTE:  
Example text for  
debug notes.

DESIGN NOTE:  
Example text for  
cautionary design  
notes.

DESIGN NOTE:  
Example text for  
critical design  
notes.

LAYOUT NOTE:  
Example text for  
critical layout  
guidelines.

## NOTES

Comment

Not fitted components are marked as **X**

**DRAFT** - Very early stage of schematic, ignore details.

**PRELIMINARY** - Close to final schematic.

**CHECKED** - There shouldn't be any mistakes. Contact the engineer if you find any.

**RELEASED** - A board with this schematic has been sent to production.

### Block Diagram

File: Block Diagram.kicad\_sch

### Revision History

File: Revision History.kicad\_sch

### Project Architecture

File: Project Architecture.kicad\_sch

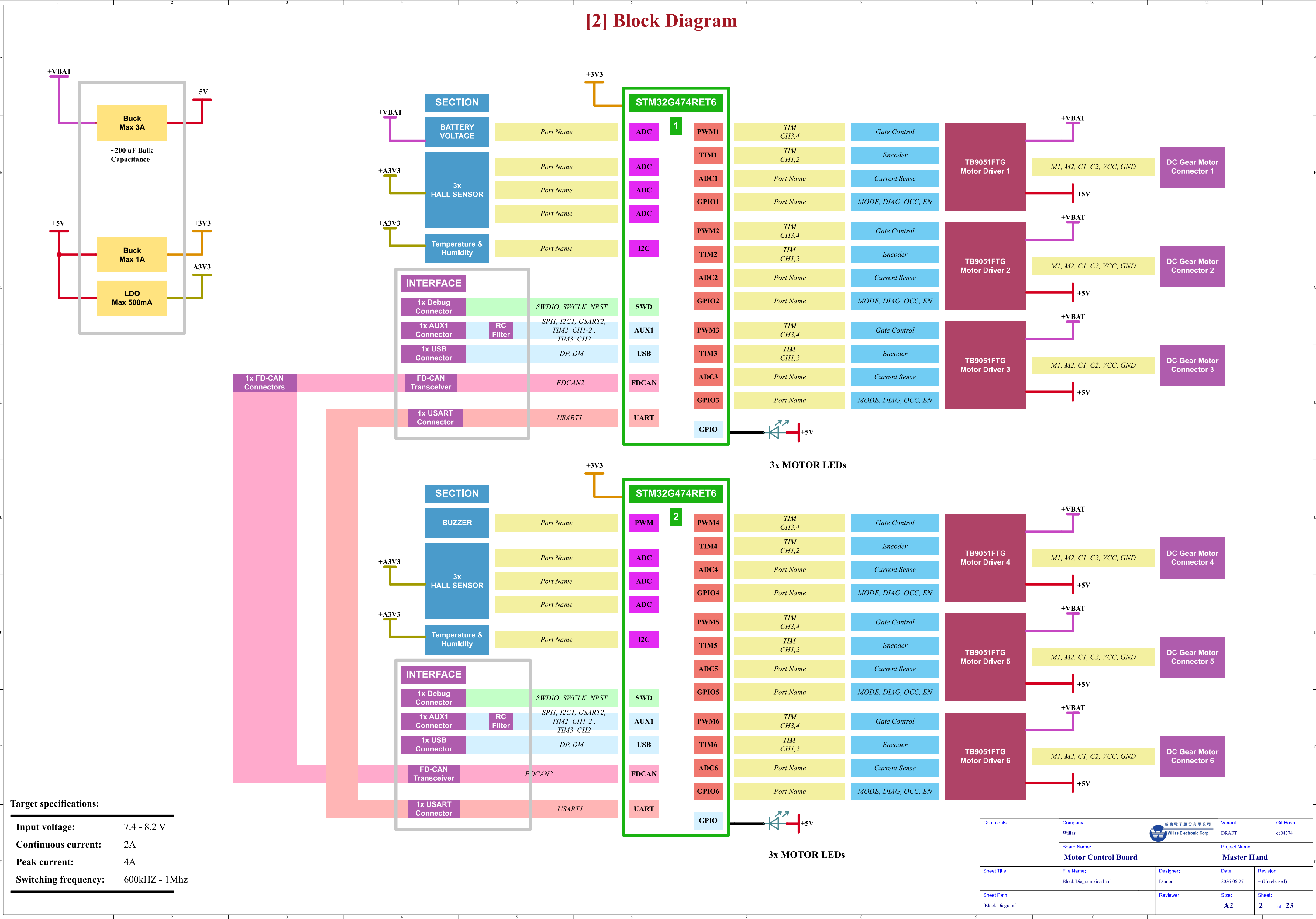
### Power - Sequencing

File: Power - Sequencing.kicad\_sch

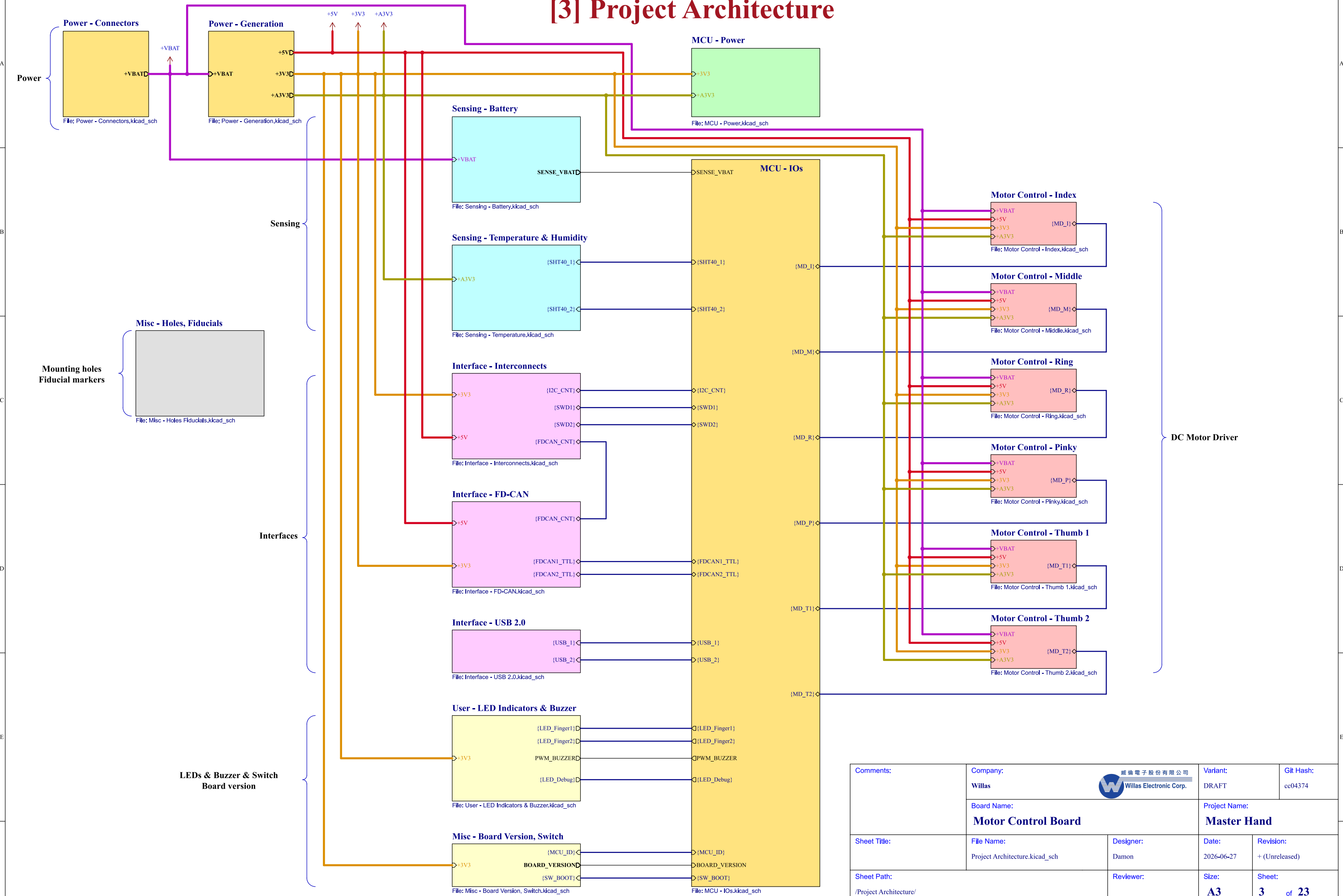
Date: 27-Aug-2026

|                  |                                           |  |                                                           |                              |                             |
|------------------|-------------------------------------------|--|-----------------------------------------------------------|------------------------------|-----------------------------|
| Comments:        | Company:<br>Willas                        |  | <div><div>W</div><div>Willas Electronic Corp.</div></div> | Variant:<br>DRAFT            | Git Hash:<br>cc04374        |
|                  | Board Name:<br>Motor Control Board        |  |                                                           | Project Name:<br>Master Hand |                             |
| Sheet Title:     | File Name:<br>Master_Hand_Kicad.kicad_sch |  | Designer:<br>Damon                                        | Date:<br>2026-06-27          | Revision:<br>+ (Unreleased) |
| Sheet Path:<br>/ |                                           |  | Reviewer:                                                 | Size:<br>A3                  | Sheet:<br>1 of 23           |

[2] Block Diagram

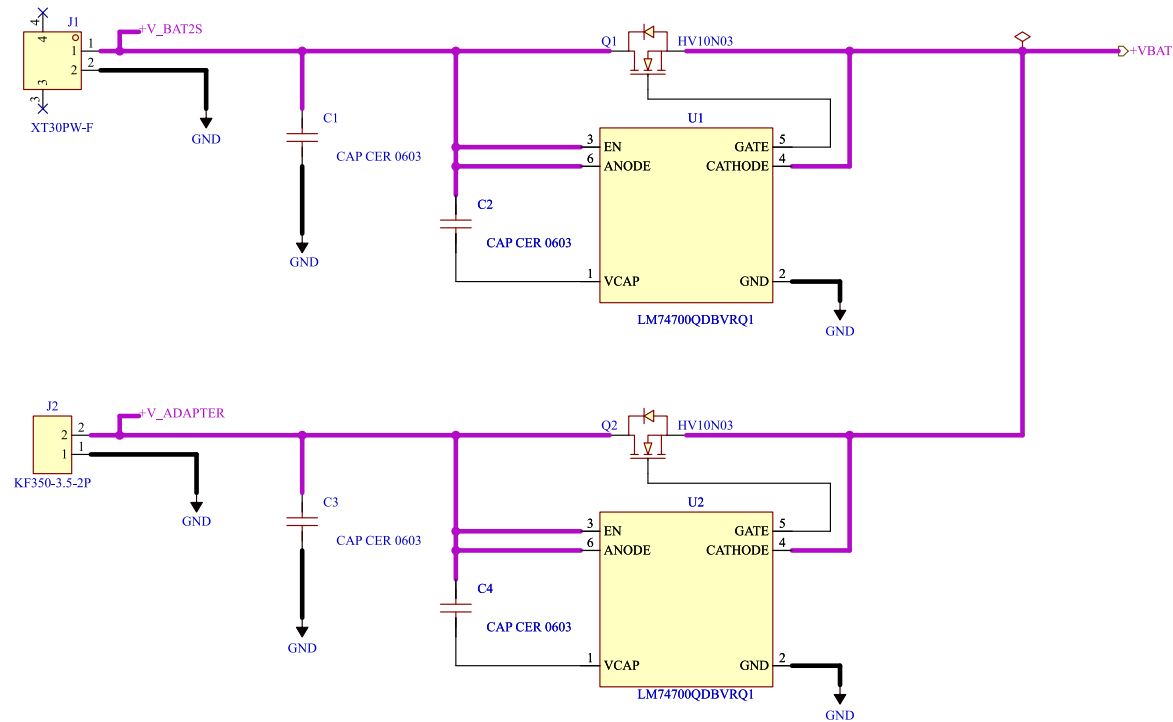
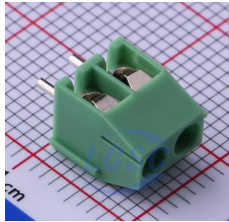


# [3] Project Architecture



|                                       |                                              |                    |                                                                                                               |                              |                      |
|---------------------------------------|----------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|
| Comments:                             | Company:<br>Willas                           |                    |  Willas Electronic Corp. | Variant:<br>DRAFT            | Git Hash:<br>cc04374 |
|                                       | Board Name:<br>Motor Control Board           |                    |                                                                                                               | Project Name:<br>Master Hand |                      |
| Sheet Title:                          | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Damon | Date:<br>2026-06-27                                                                                           | Revision:<br>+ (Unreleased)  |                      |
| Sheet Path:<br>/Project Architecture/ |                                              | Reviewer:          | Size:<br>A3                                                                                                   | Sheet:<br>3 of 23            |                      |

## [4] Power - Connectors

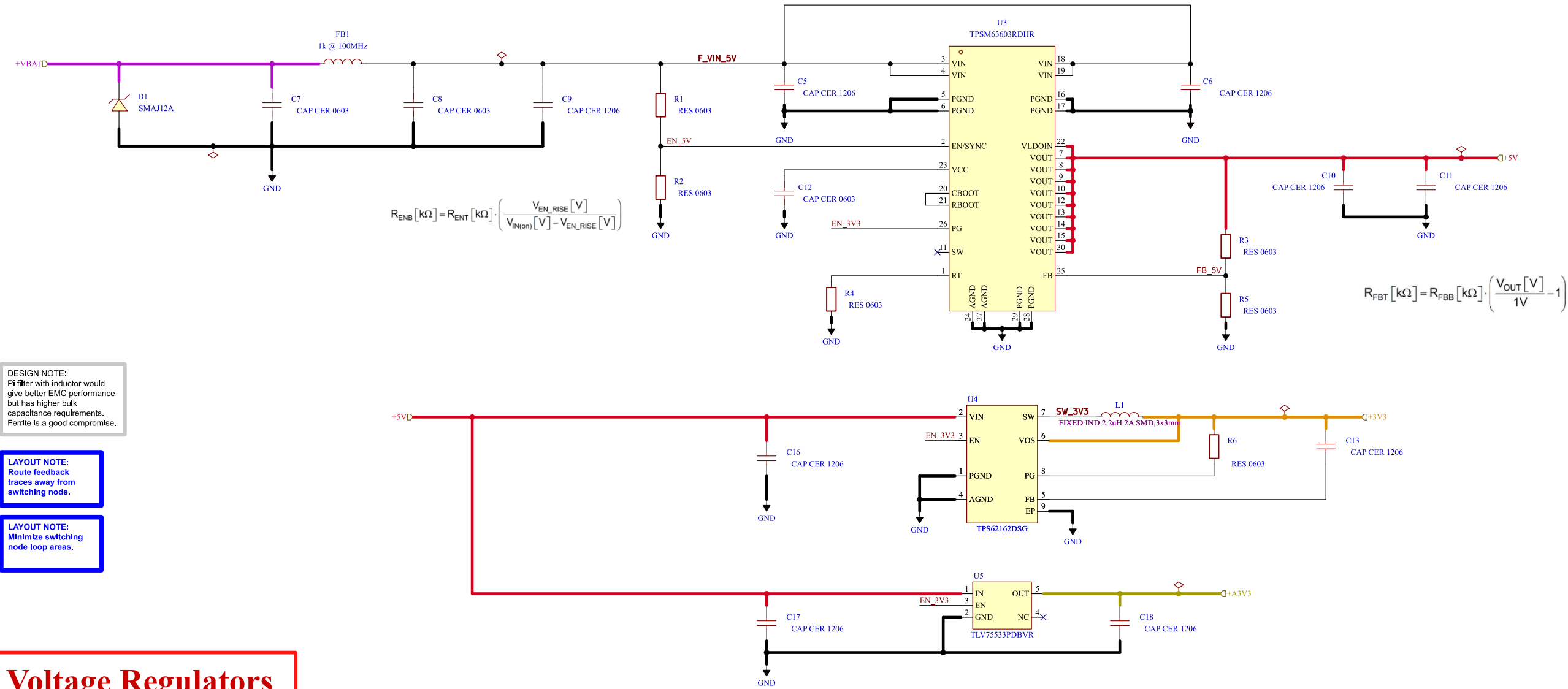


## Power Connectors - Path Control

|                                                          |                                                                                                                                                                                |  |                    |                                     |                                 |  |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------|-------------------------------------|---------------------------------|--|
| Comments:                                                | Company:<br><div><div>Willas</div><div>威倫電子股份有限公司<br/>Willas Electronic Corp.</div></div> |  | Variant:<br>DRAFT  |                                     | Git Hash:<br>cc04374            |  |
|                                                          | Board Name:<br><b>Motor Control Board</b>                                                                                                                                      |  |                    | Project Name:<br><b>Master Hand</b> |                                 |  |
| Sheet Title:                                             | File Name:<br>Power - Connectors.kicad_sch                                                                                                                                     |  | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased)     |  |
| Sheet Path:<br>/Project Architecture/Power - Connectors/ |                                                                                                                                                                                |  | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>4</b> of <b>23</b> |  |



[5] Power - Generation



Voltage Regulators

DC BULK

DC-Link Bulk Capacitors. 330uF + 10 x 10 uF = 430uF (35V)

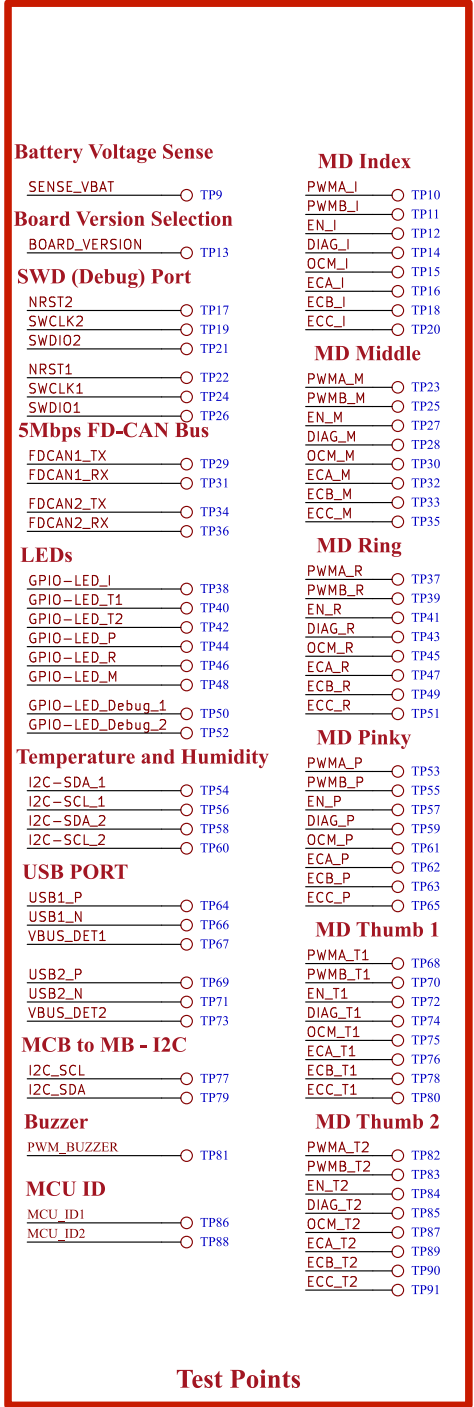
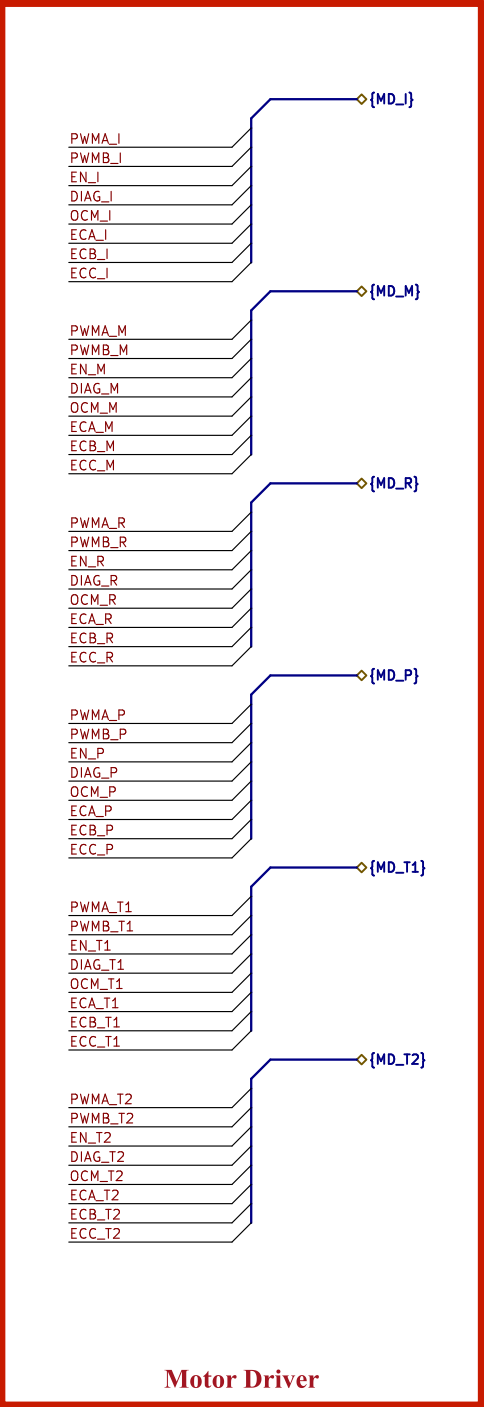
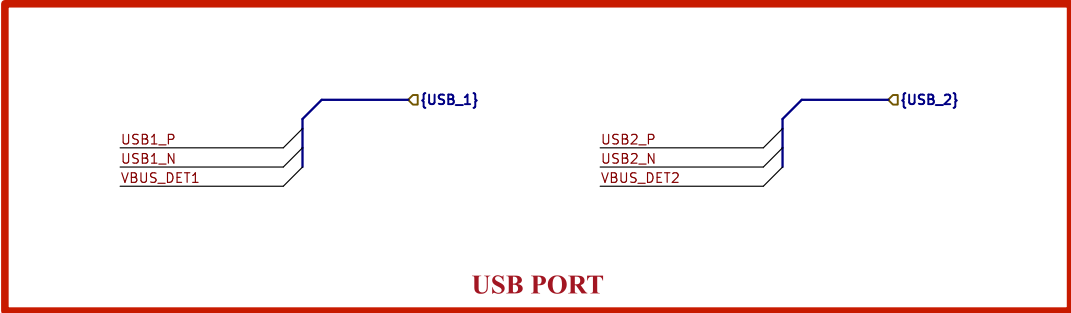
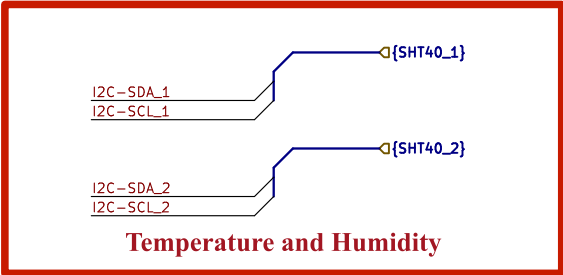
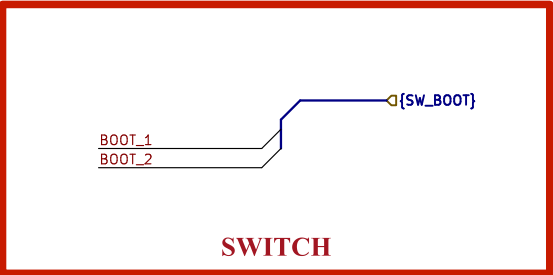
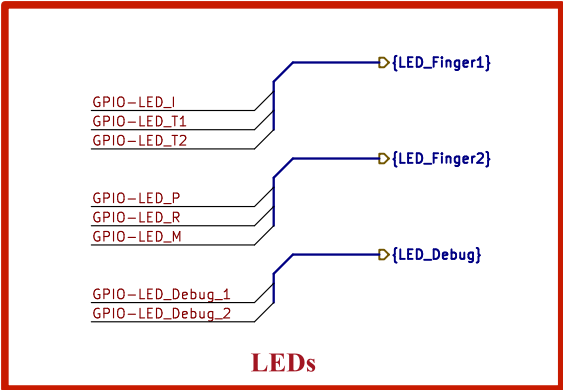
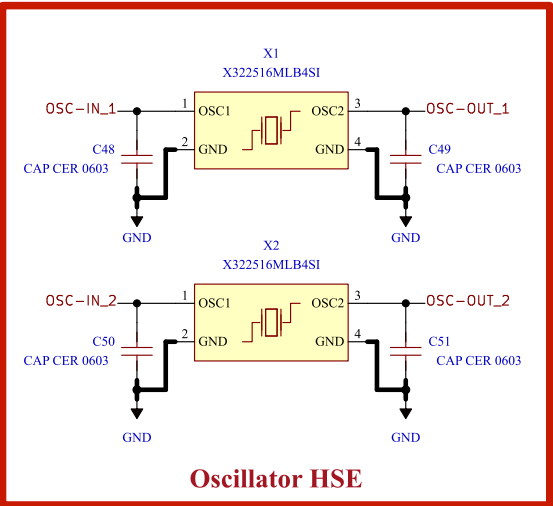
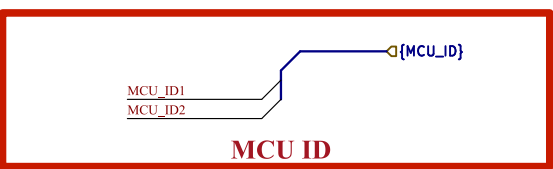
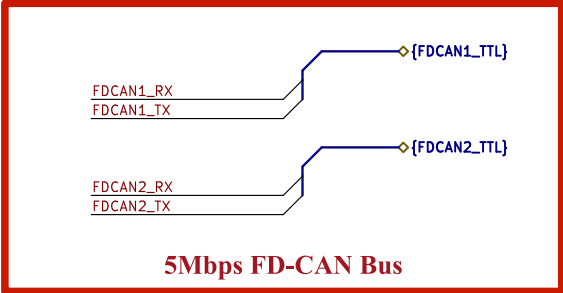
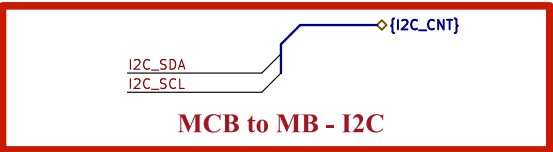
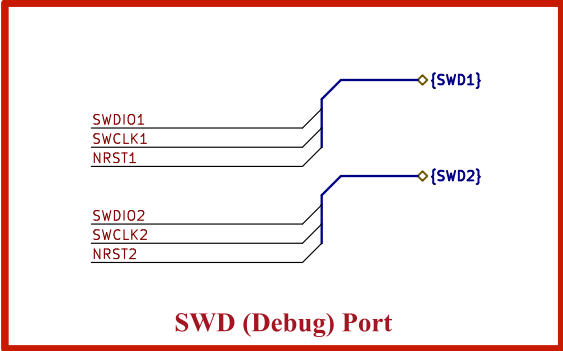
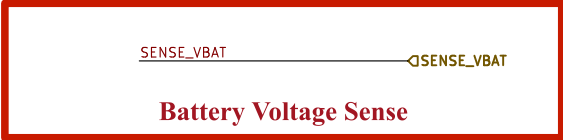
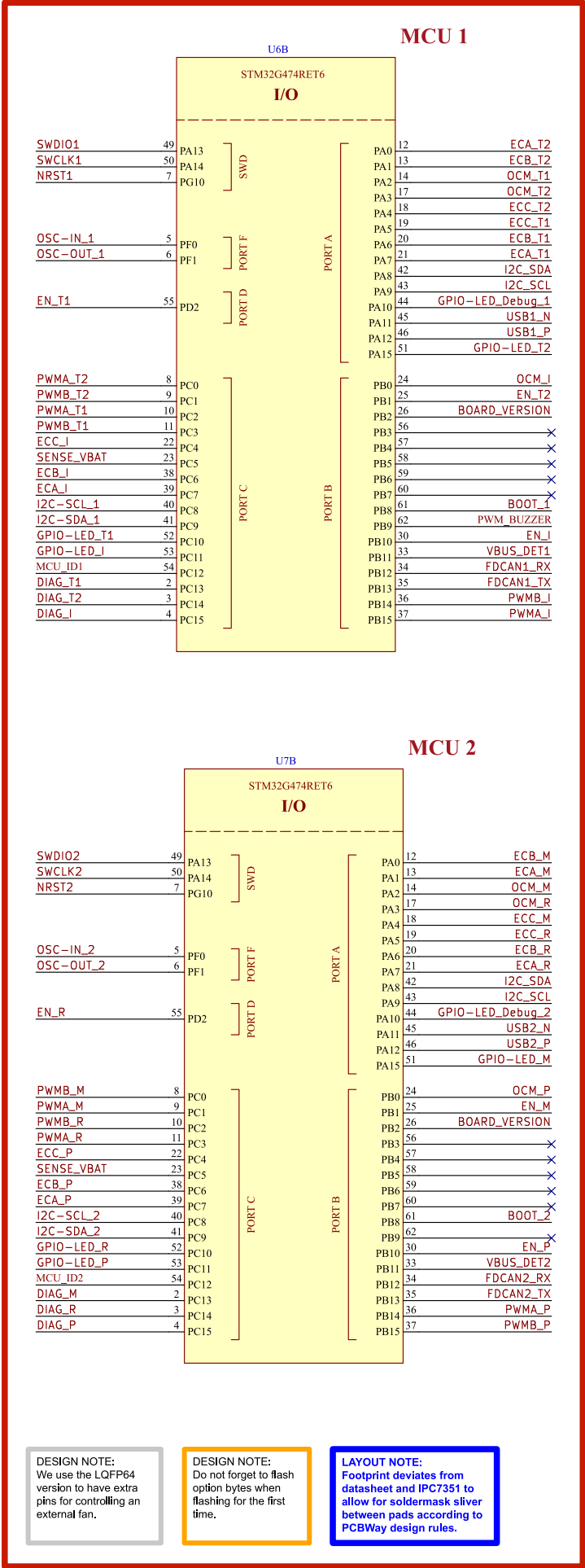
Test Points



|                                                       |                                                    |                 |                           |                          |
|-------------------------------------------------------|----------------------------------------------------|-----------------|---------------------------|--------------------------|
| Comments:                                             | Company: Willas 威倫電子股份有限公司 Willas Electronic Corp. |                 | Variant: DRAFT            | Git Hash: cc04374        |
|                                                       | Board Name: Motor Control Board                    |                 | Project Name: Master Hand |                          |
| Sheet Title:                                          | File Name: Power - Generation.kicad_sch            | Designer: Damon | Date: 2026-06-27          | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Power - Generation/ | Reviewer:                                          | Size: A3        | Sheet: 5 of 23            |                          |

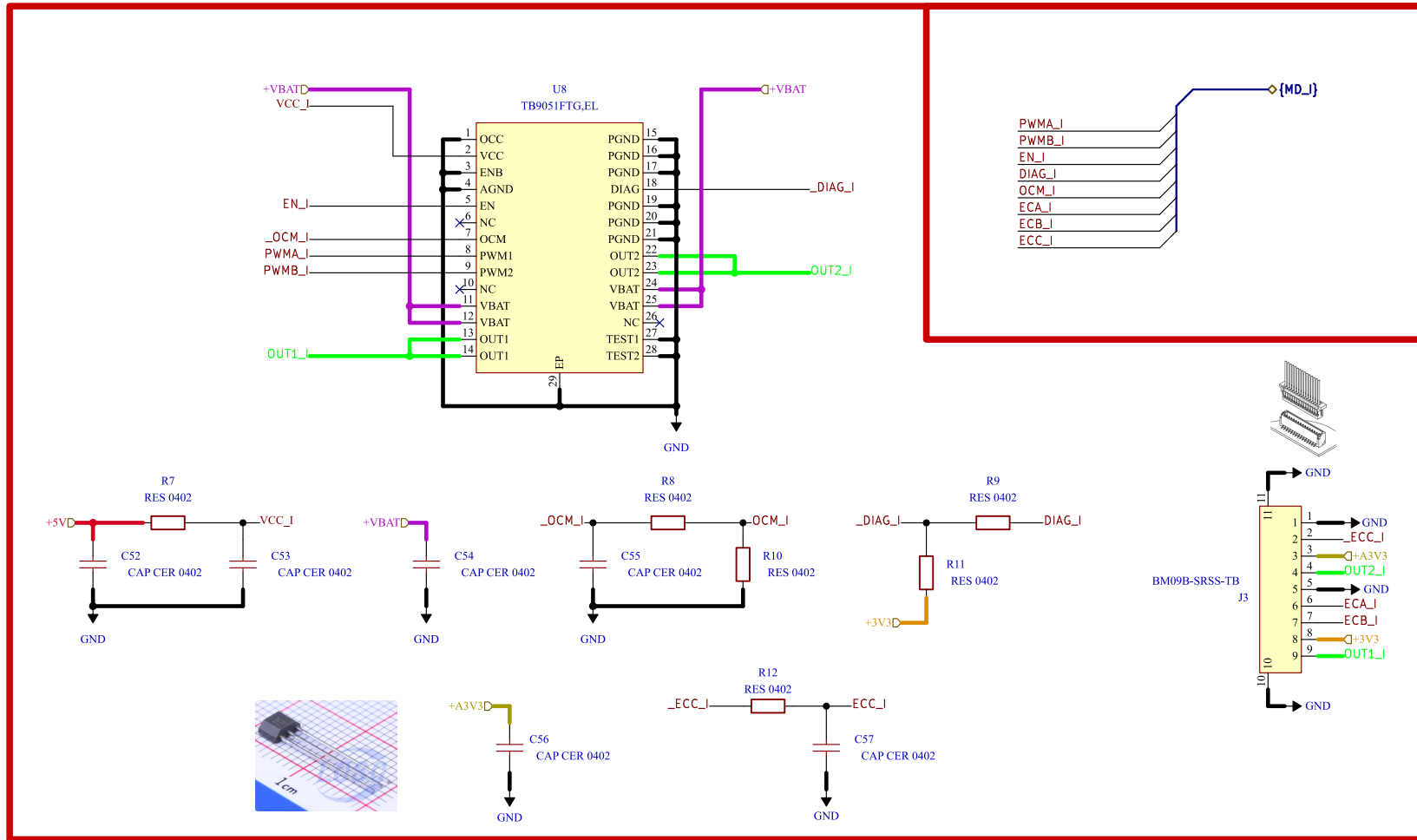


[7] MCU - IOs



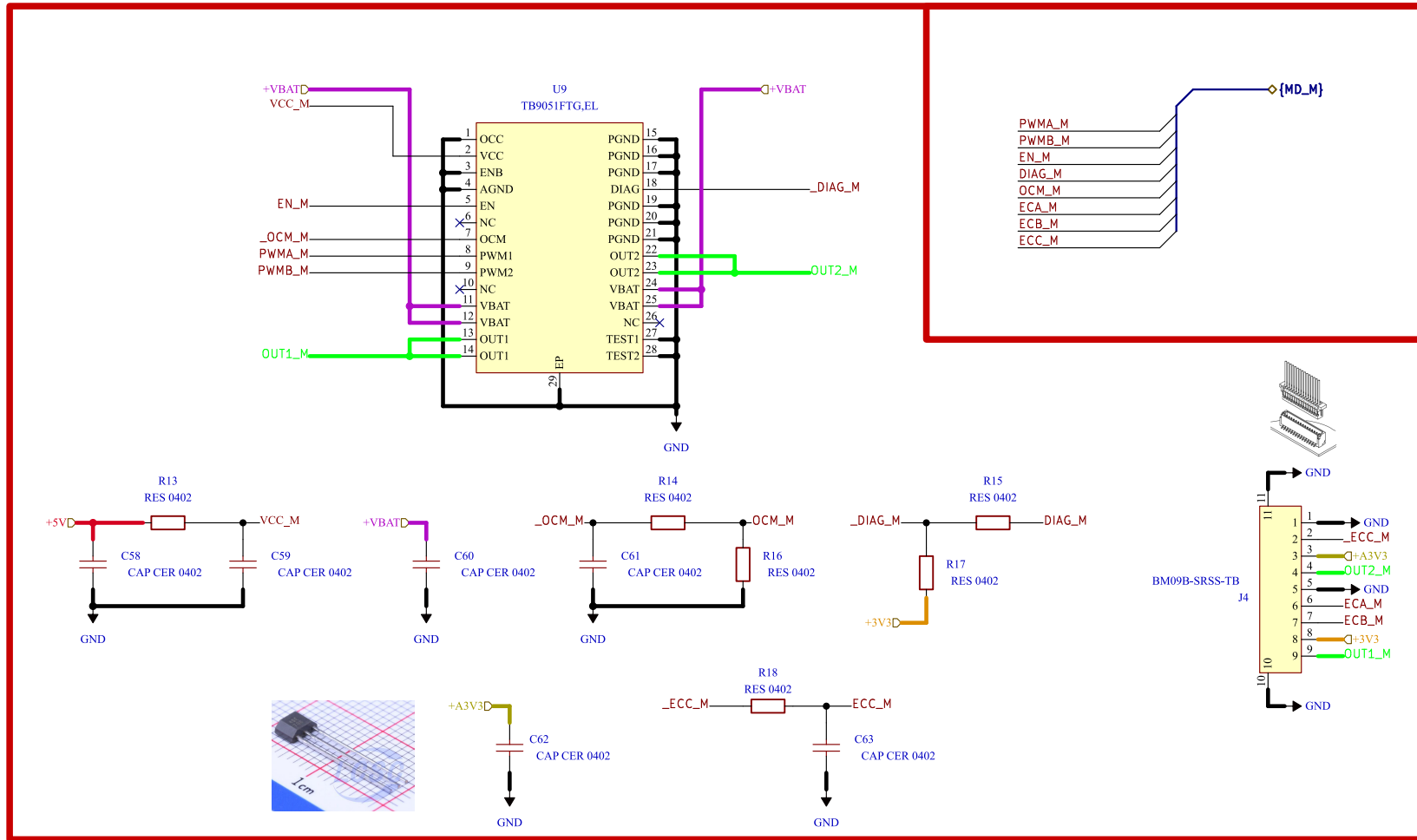
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|-----------------------------------------------------|-------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------|
| Comments:                                           | Company:<br><br>Willas                    |                        | <br>威倫電子股份有限公司<br>Willas Electronic Corp. | Variant:<br><br>DRAFT               | Git Hash:<br><br>cc04374 |
|                                                     | Board Name:<br><b>Motor Control Board</b> |                        |                                                                                                                                | Project Name:<br><b>Master Hand</b> |                          |
| Sheet Title:                                        | File Name:<br><br>MCU - IOs.kicad_sch     | Designer:<br><br>Damon | Date:<br><br>2026-06-27                                                                                                        | Revision:<br><br>+ (Unreleased)     |                          |
| Sheet Path:<br><br>/Project Architecture/MCU - IOs/ |                                           | Reviewer:              | Size:<br><br><b>A3</b>                                                                                                         | Sheet:<br><br><b>7</b> of <b>23</b> |                          |

# [8] Motor Control - Index



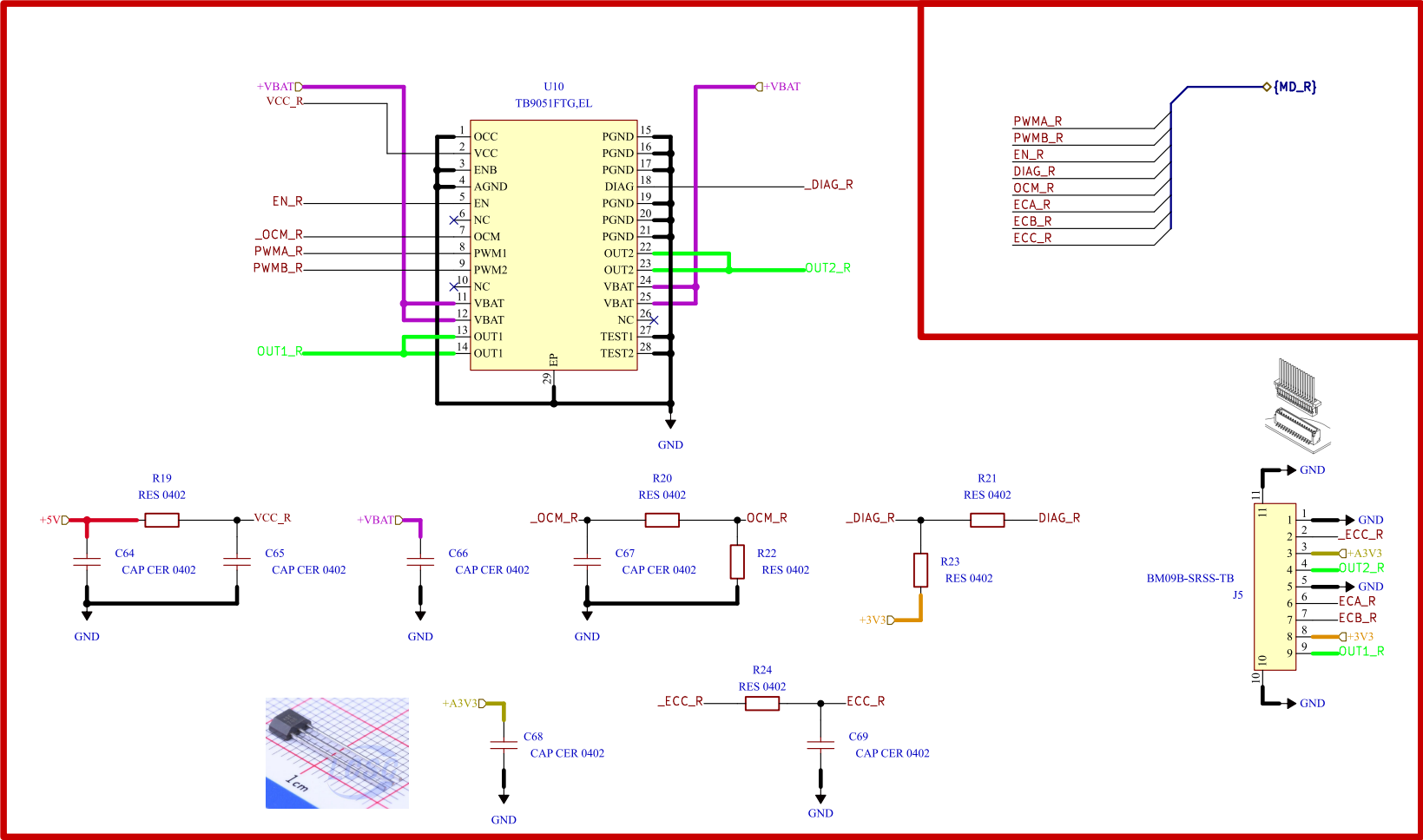
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| Comments:                                                   | Company:<br><div>Willas</div> <div>Willas Electronic Corp.</div> |                    | Variant:<br>DRAFT                   | Git Hash:<br>cc04374            |
|                                                             | Board Name:<br><b>Motor Control Board</b>                                                                                                             |                    | Project Name:<br><b>Master Hand</b> |                                 |
| Sheet Title:                                                | File Name:<br>Motor Control - Index.kicad_sch                                                                                                         | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased)     |
| Sheet Path:<br>/Project Architecture/Motor Control - Index/ |                                                                                                                                                       | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>8</b> of <b>23</b> |

# [9] Motor Control - Middle



|                                                              |                                                                                                                                                                                |                    |                                     |                                 |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------|---------------------------------|
| Comments:                                                    | Company:<br><div><div>Willas</div><div>威倫電子股份有限公司<br/>Willas Electronic Corp.</div></div> |                    | Variant:<br>DRAFT                   | Git Hash:<br>cc04374            |
|                                                              | Board Name:<br><b>Motor Control Board</b>                                                                                                                                      |                    | Project Name:<br><b>Master Hand</b> |                                 |
| Sheet Title:                                                 | File Name:<br>Motor Control - Middle.kicad_sch                                                                                                                                 | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased)     |
| Sheet Path:<br>/Project Architecture/Motor Control - Middle/ |                                                                                                                                                                                | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>9</b> of <b>23</b> |

# [10] Motor Control - Ring



Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

cc04374

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Motor Control - Ring.kicad\_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Motor Control - Ring/

Reviewer:

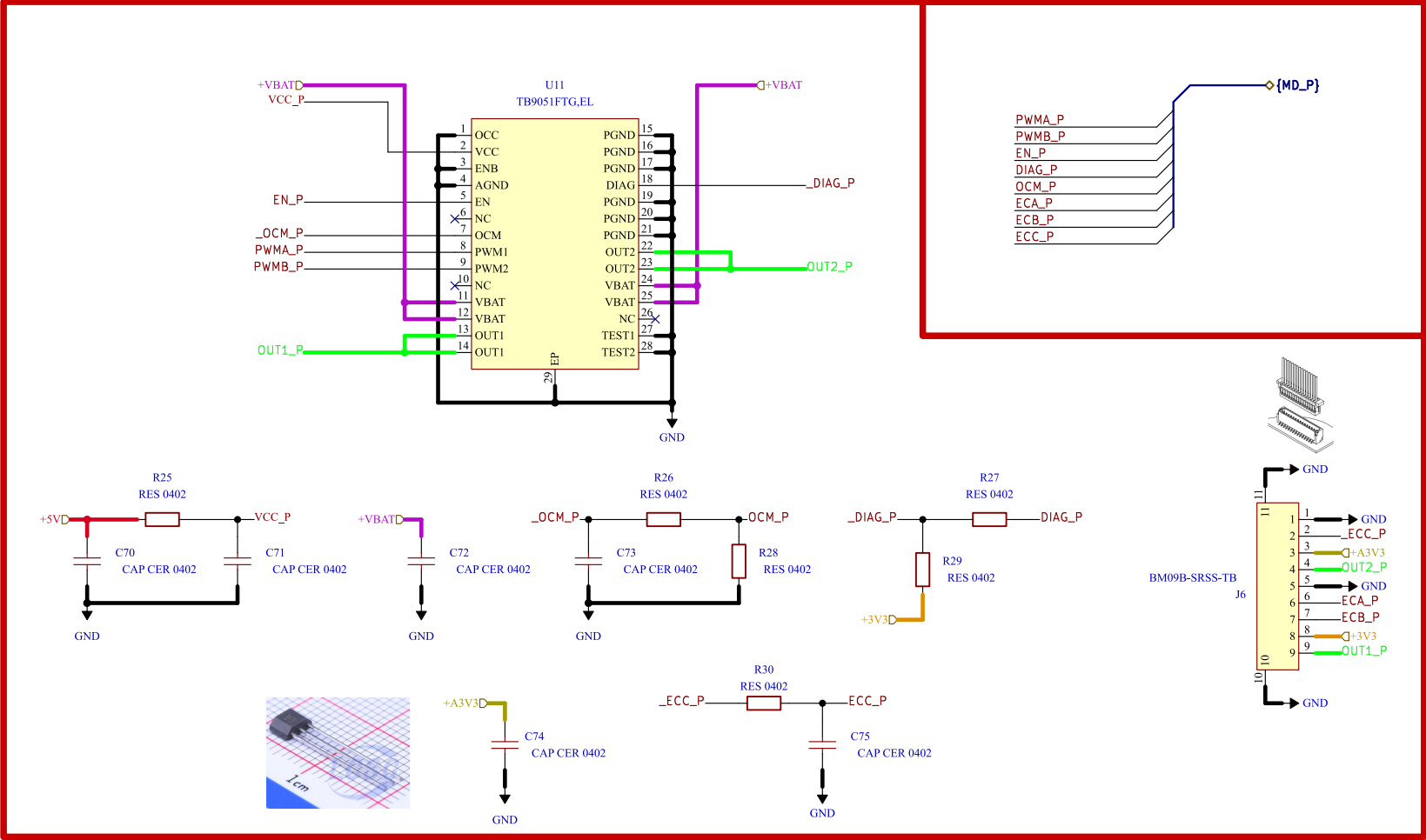
Size:

A4

Sheet:

10 of 23

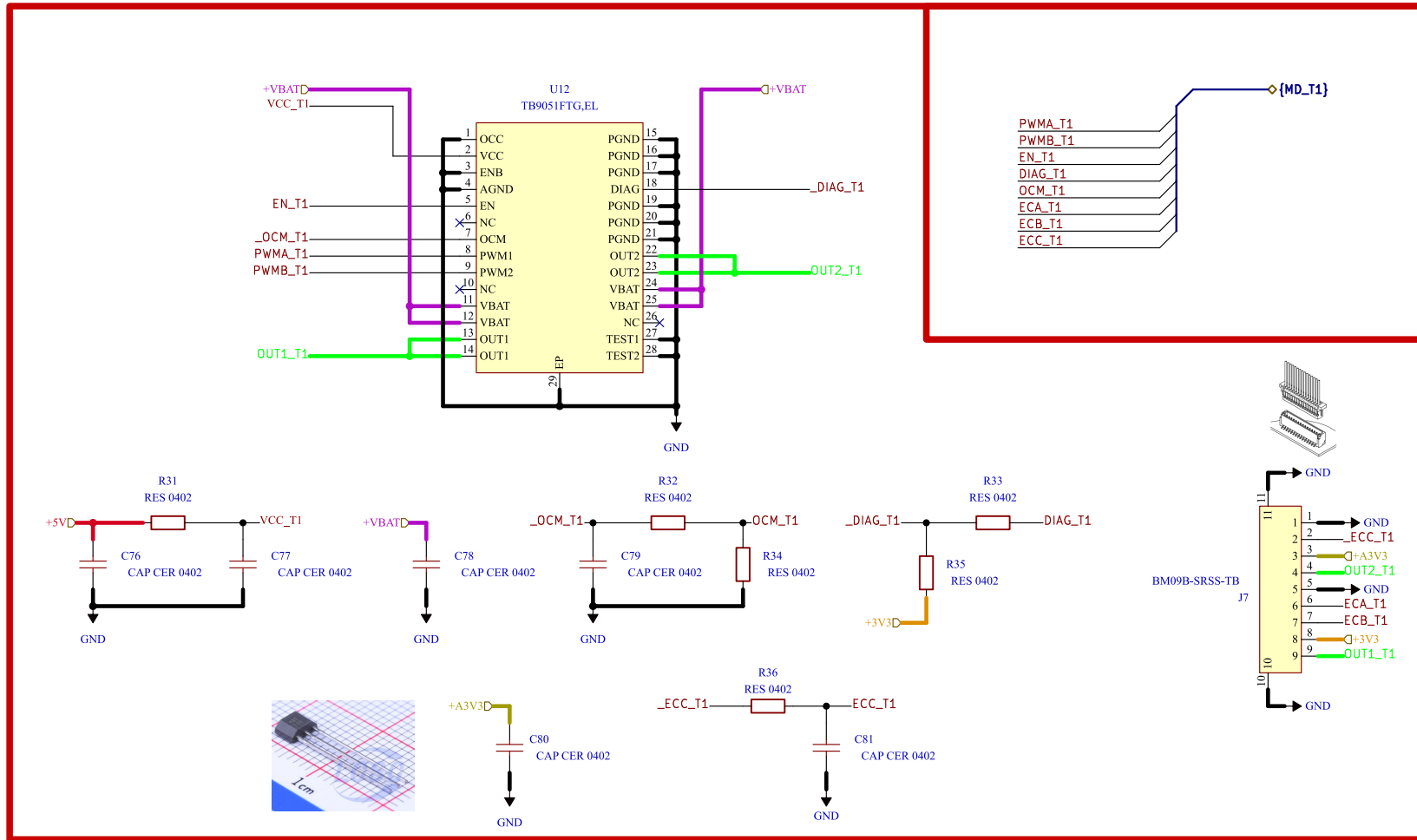
[11] Motor Control - Pinky



|                                                             |                                                                                                                                                                             |                    |                                     |                                  |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------|----------------------------------|
| Comments:                                                   | Company:<br><div>威倫電子股份有限公司</div> <div>Willas</div> <div>Willas Electronic Corp.</div> |                    | Variant:<br>DRAFT                   | Git Hash:<br>cc04374             |
|                                                             | Board Name:<br><b>Motor Control Board</b>                                                                                                                                   |                    | Project Name:<br><b>Master Hand</b> |                                  |
| Sheet Title:                                                | File Name:<br>Motor Control - Pinky.kicad_sch                                                                                                                               | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased)      |
| Sheet Path:<br>/Project Architecture/Motor Control - Pinky/ |                                                                                                                                                                             | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>11</b> of <b>23</b> |

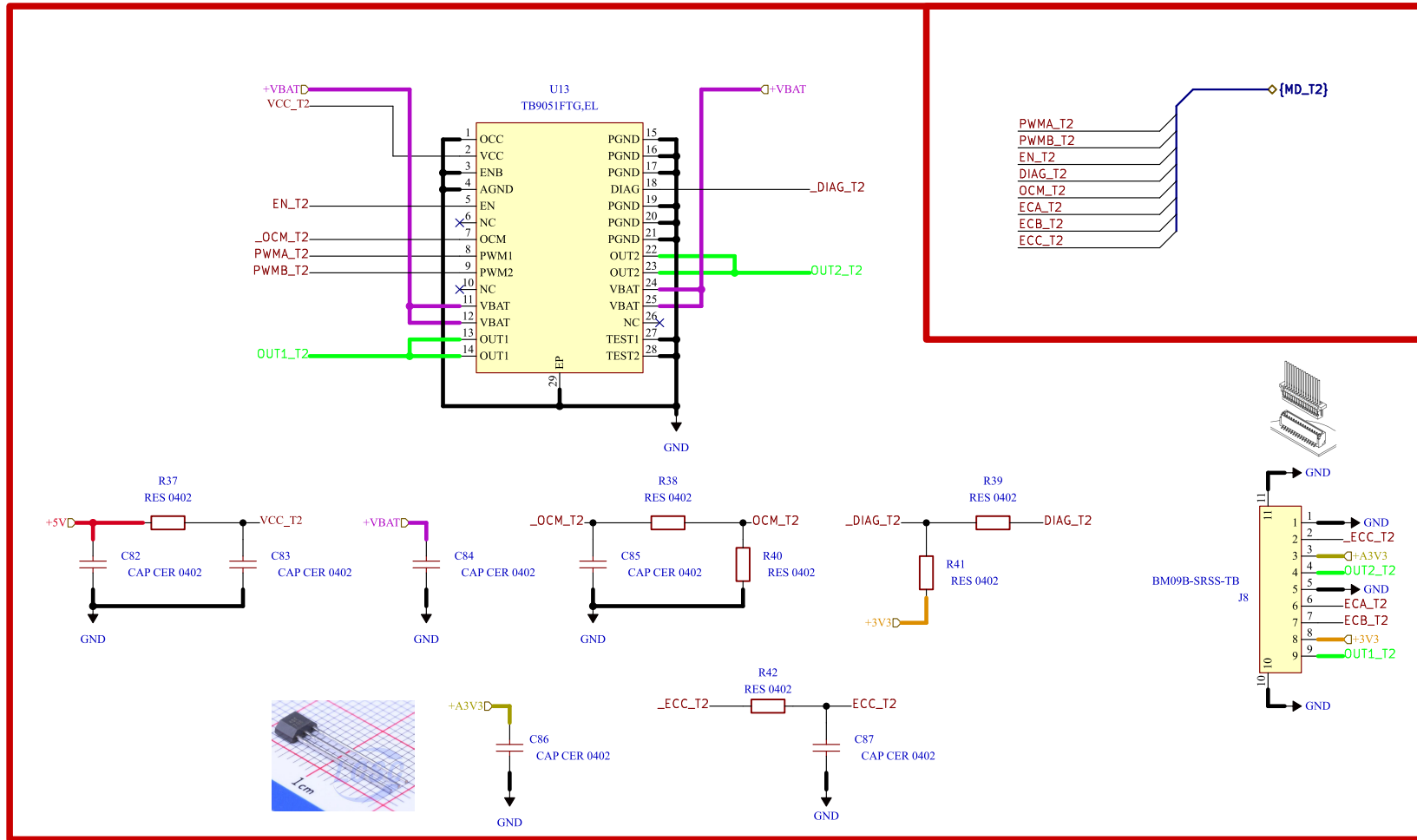


# [12] Motor Control - Thumb 1



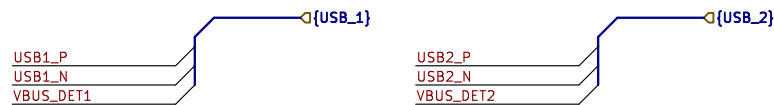
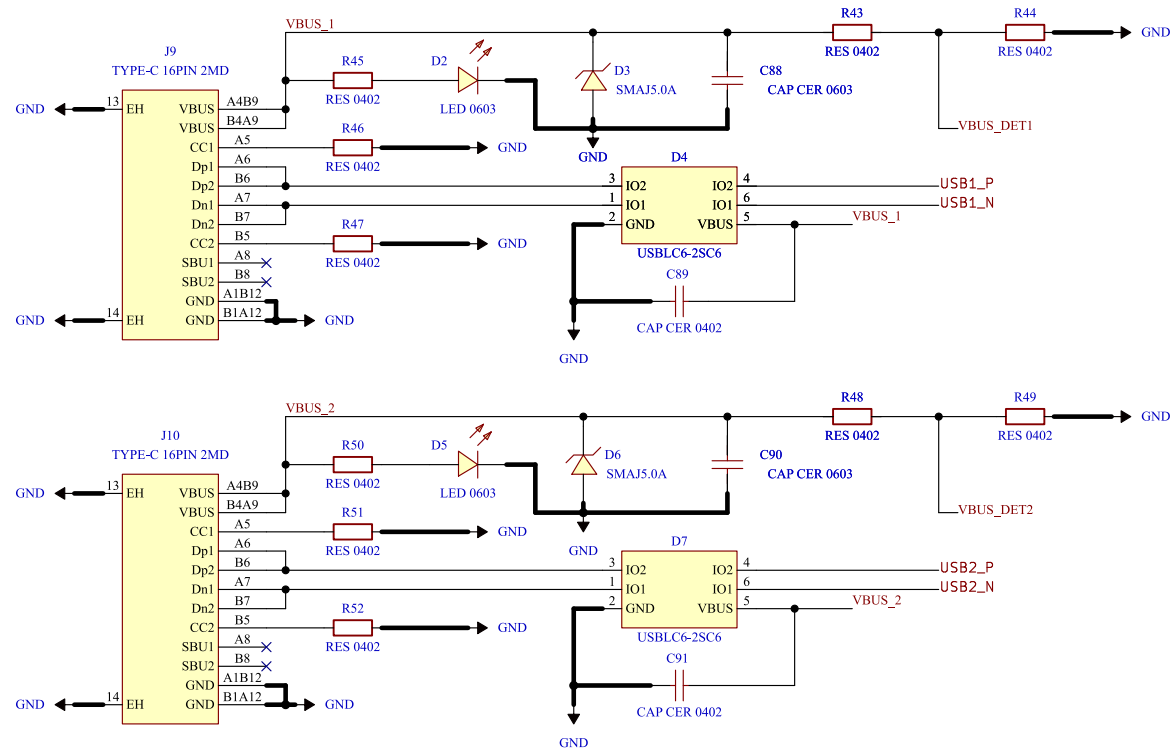
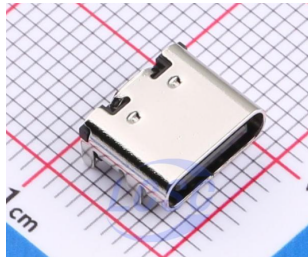
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|------------------------------------------------------------|----------------------------------------------|-----------------|---------------------------|--------------------------|
| Comments:                                                  | Company: 威倫電子股份有限公司<br>Willas                |                 | Variant: DRAFT            | Git Hash: cc04374        |
|                                                            | Board Name: Motor Control Board              |                 | Project Name: Master Hand |                          |
| Sheet Title:                                               | File Name: Motor Control - Thumb 1.kicad_sch | Designer: Damon | Date: 2026-06-27          | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Motor Control - Thumb 1/ |                                              | Reviewer:       | Size: A4                  | Sheet: 12 of 23          |

# [13] Motor Control - Thumb 2



|                                                               |                                                                                                                                                                                |                    |                              |                             |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------|-----------------------------|
| Comments:                                                     | Company:<br><div><div>Willas</div><div>威倫電子股份有限公司<br/>Willas Electronic Corp.</div></div> |                    | Variant:<br>DRAFT            | Git Hash:<br>cc04374        |
|                                                               | Board Name:<br>Motor Control Board                                                                                                                                             |                    | Project Name:<br>Master Hand |                             |
| Sheet Title:                                                  | File Name:<br>Motor Control - Thumb 2.kicad_sch                                                                                                                                | Designer:<br>Damon | Date:<br>2026-06-27          | Revision:<br>+ (Unreleased) |
| Sheet Path:<br>/Project Architecture/Motor Control - Thumb 2/ |                                                                                                                                                                                | Reviewer:          | Size:<br>A4                  | Sheet:<br>13 of 23          |

# [14] Interface - USB 2.0



## DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS\_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB\_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPU1 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

cc04374

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad\_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

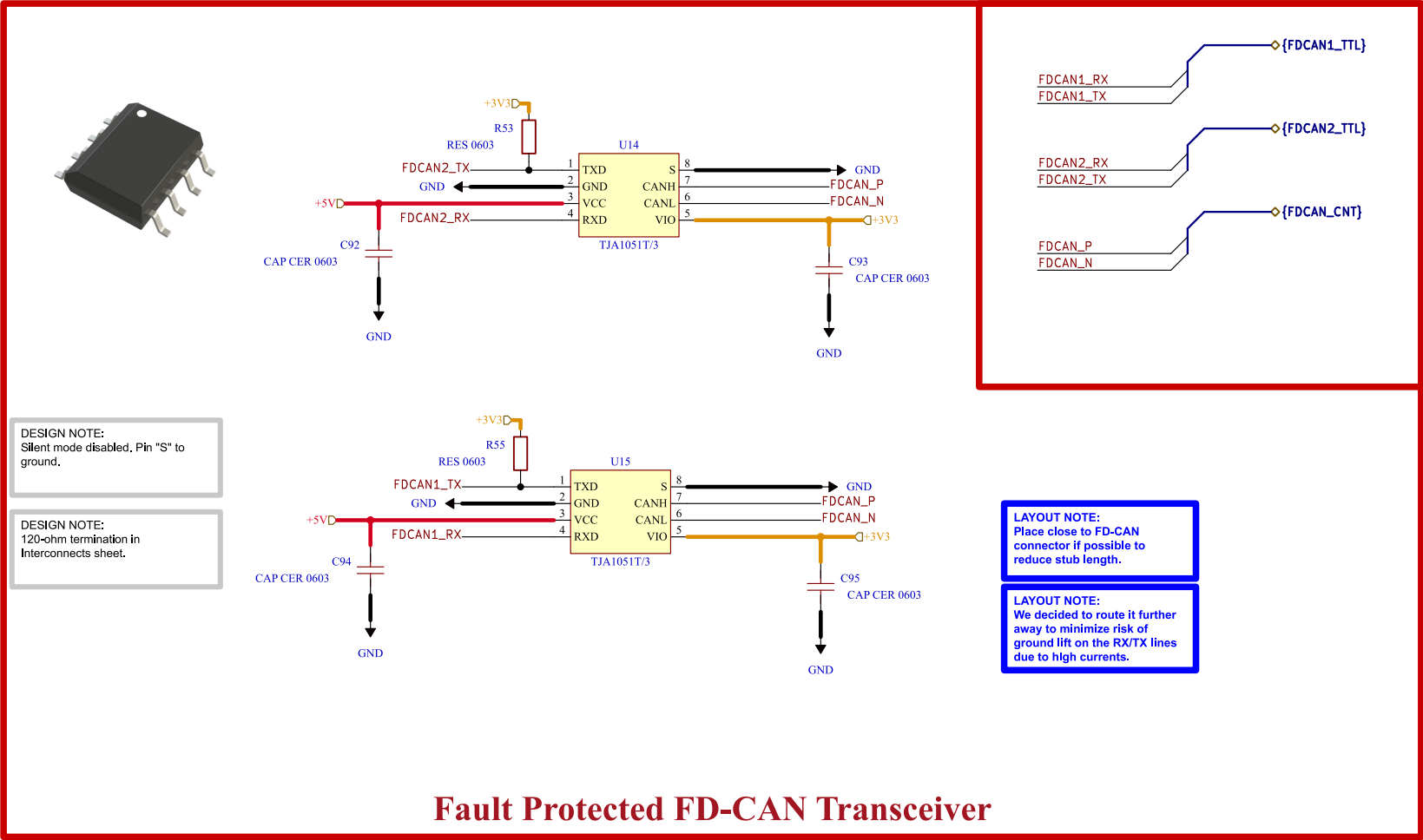
Size:

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Sheet:

14 of 23

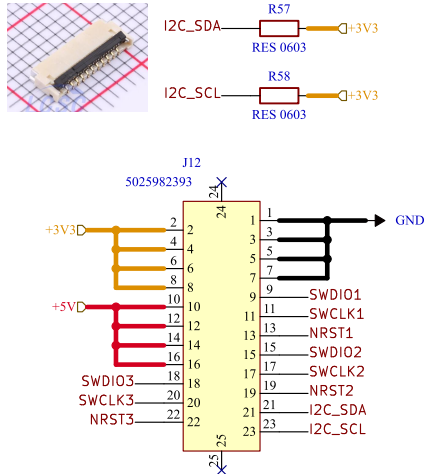
[15] Interface - FD-CAN



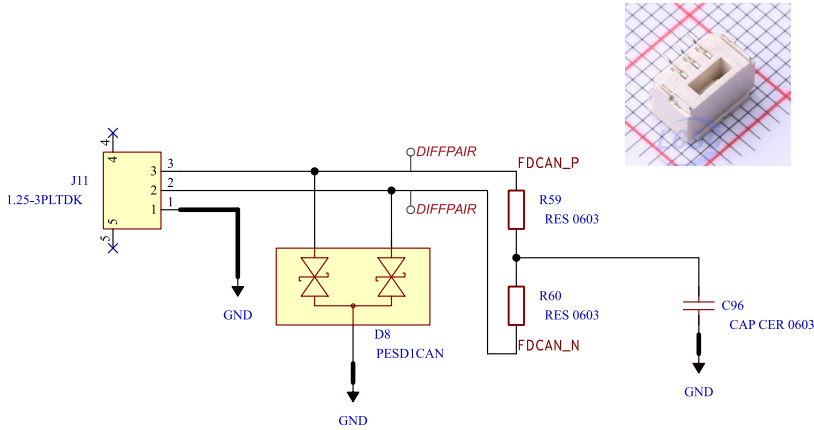
Fault Protected FD-CAN Transceiver

|                                                          |                                                                                                                                                       |                    |                     |                                     |                      |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|-------------------------------------|----------------------|
| Comments:                                                | Company:<br><div>Willas</div> <div>Willas Electronic Corp.</div> |                    | Variant:<br>DRAFT   |                                     | Git Hash:<br>cc04374 |
|                                                          | Board Name:<br><b>Motor Control Board</b>                                                                                                             |                    |                     | Project Name:<br><b>Master Hand</b> |                      |
| Sheet Title:                                             | File Name:<br>Interface - FD-CAN.kicad_sch                                                                                                            | Designer:<br>Damon | Date:<br>2026-06-27 | Revision:<br>+ (Unreleased)         |                      |
| Sheet Path:<br>/Project Architecture/Interface - FD-CAN/ |                                                                                                                                                       | Reviewer:          | Size:<br>A4         | Sheet:<br>15 of 23                  |                      |

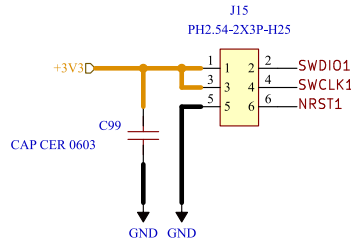
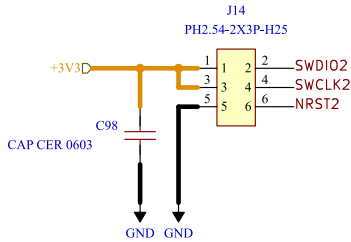
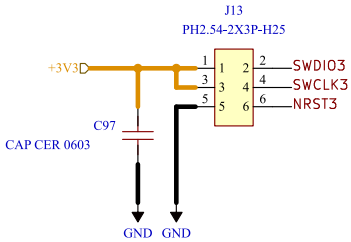
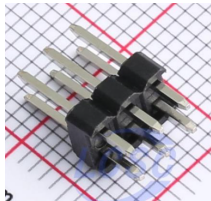
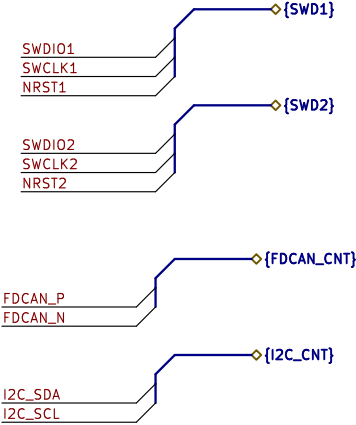
[16] Interface - Interconnects



FFC Connector MCB & MB



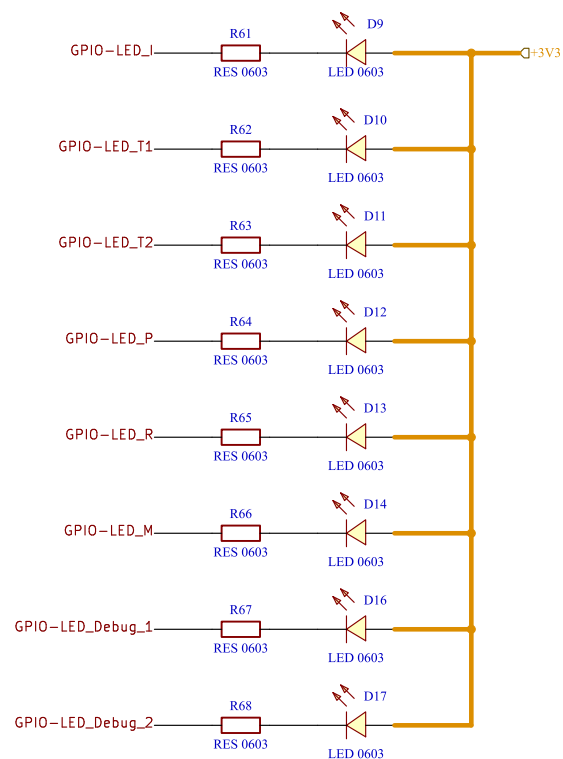
FD-CAN Connector and Termination



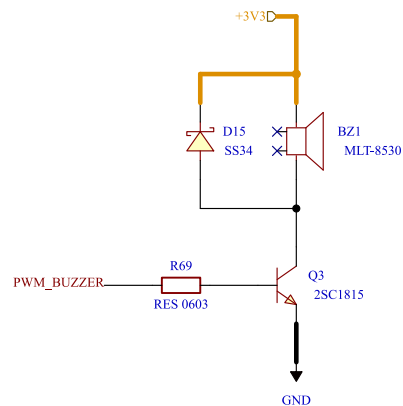
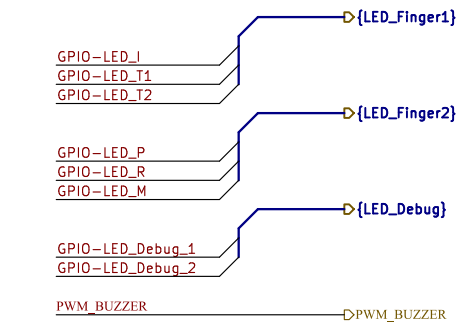
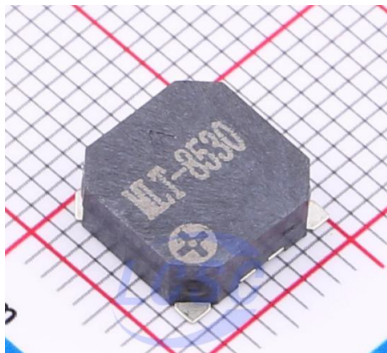
SWD Debug Connector

|                                                                 |                                                                                                                                                       |                    |                     |                                     |                      |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|-------------------------------------|----------------------|
| Comments:                                                       | Company:<br><div>Willas</div> <div>Willas Electronic Corp.</div> |                    | Variant:<br>DRAFT   |                                     | Git Hash:<br>cc04374 |
|                                                                 | Board Name:<br><b>Motor Control Board</b>                                                                                                             |                    |                     | Project Name:<br><b>Master Hand</b> |                      |
| Sheet Title:                                                    | File Name:<br>Interface - Interconnects.kicad_sch                                                                                                     | Designer:<br>Damon | Date:<br>2026-06-27 | Revision:<br>+ (Unreleased)         |                      |
| Sheet Path:<br>/Project Architecture/Interface - Interconnects/ |                                                                                                                                                       | Reviewer:          | Size:<br><b>A4</b>  | Sheet:<br><b>16</b> of <b>23</b>    |                      |

# [17] User - LED Indicators & Buzzer



| LED Color | Typical Vf Range |
|-----------|------------------|
| Red       | 1.8 to 2.1       |
| Amber     | 2 to 2.2         |
| Orange    | 1.9 to 2.2       |
| Yellow    | 1.9 to 2.2       |
| Green     | 2 to 3.1         |
| Blue      | 3 to 3.7         |
| White     | 3 to 3.4         |



Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

cc04374

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

User - LED Indicators & Buzzer.kicad\_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/User - LED Indicators & Buzzer/

Reviewer:

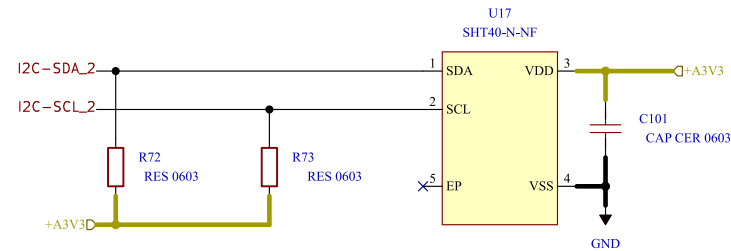
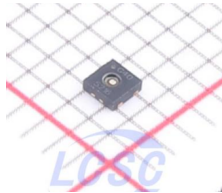
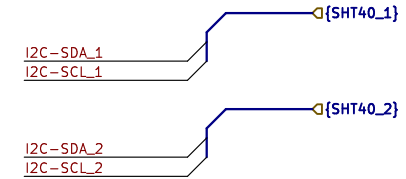
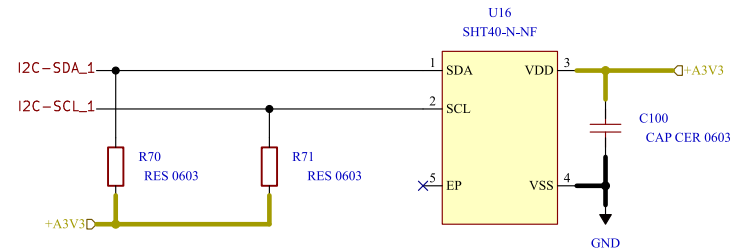
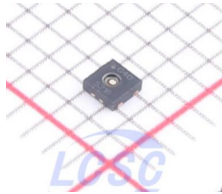
Size:

A4

Sheet:

17 of 23

# [18] Sensing - Temperature & Humidity



|                                                                        |                                                                                                                                                                                |                    |                                     |                                  |                      |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------|----------------------------------|----------------------|
| Comments:                                                              | Company:<br><div><div>Willas</div><div>威倫電子股份有限公司<br/>Willas Electronic Corp.</div></div> |                    | Variant:<br>DRAFT                   |                                  | Git Hash:<br>cc04374 |
|                                                                        | Board Name:<br><b>Motor Control Board</b>                                                                                                                                      |                    | Project Name:<br><b>Master Hand</b> |                                  |                      |
| Sheet Title:                                                           | File Name:<br>Sensing - Temperature.kicad_sch                                                                                                                                  | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased)      |                      |
| Sheet Path:<br>/Project Architecture/Sensing - Temperature & Humidity/ |                                                                                                                                                                                | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>18</b> of <b>23</b> |                      |

# [19] Sensing - Battery

**DESIGN NOTE:**  
Battery voltage sensing divider for 2S LiPo battery.

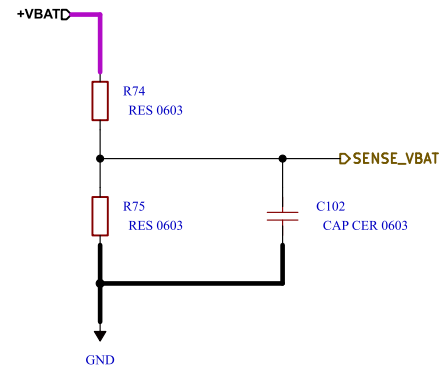
R1 = 100kΩ (Top)  
R2 = 47kΩ (Bottom)

**Divider Ratio:**  
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$   
 $V_{ADC} = V_{BAT} \times 0.3197$

**2S LiPo:**  
Max battery voltage (8.4V): 2.69 V  
Nominal voltage (7.4V): 2.37 V  
Low battery voltage (6.4V): 2.05 V  
Cutoff voltage (6.0V): 1.92 V

**Maximum divider current:**  
 $I = 8.4V / 147k\Omega = 57 \mu A$

**Notes:**  
- Designed for 3.3V ADC input.  
- Add 100nF capacitor at ADC input for noise filtering.  
- Use 1% resistors or better.  
- Use low temperature coefficient resistors ( $\leq 100 \text{ ppm}/^\circ\text{C}$ ) for improved measurement accuracy.



## Battery Voltage Sensing

|                                                         |                                                                                                          |                    |                                     |                             |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------|-----------------------------|
| Comments:                                               | Company:<br>Willas  |                    | Variant:<br>DRAFT                   | Git Hash:<br>cc04374        |
|                                                         | Board Name:<br><b>Motor Control Board</b>                                                                |                    | Project Name:<br><b>Master Hand</b> |                             |
| Sheet Title:                                            | File Name:<br>Sensing - Battery.kicad_sch                                                                | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased) |
| Sheet Path:<br>/Project Architecture/Sensing - Battery/ |                                                                                                          | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>19 of 23</b>   |



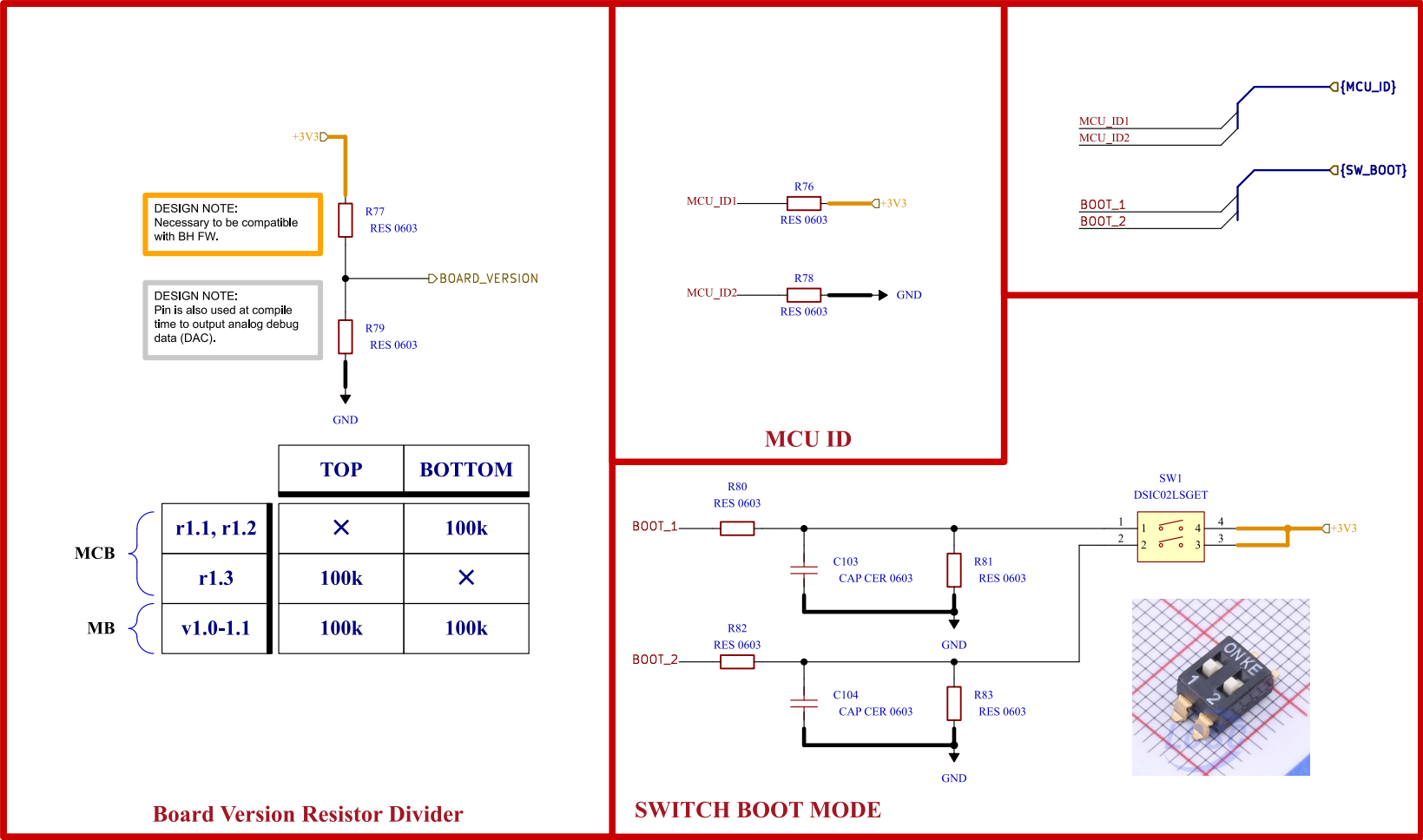
**[20] Misc - Holes, Fiducials**

**Mounting Holes and Fiducial Markers**

|                                                               |                                                                                                             |                    |                                     |                                  |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------|----------------------------------|
| Comments:                                                     | Company:<br>Willas<br> |                    | Variant:<br>DRAFT                   | Git Hash:<br>cc04374             |
|                                                               | Board Name:<br><b>Motor Control Board</b>                                                                   |                    | Project Name:<br><b>Master Hand</b> |                                  |
| Sheet Title:                                                  | File Name:<br>Misc - Holes Fiducials.kicad_sch                                                              | Designer:<br>Damon | Date:<br>2026-06-27                 | Revision:<br>+ (Unreleased)      |
| Sheet Path:<br>/Project Architecture/Misc - Holes, Fiducials/ |                                                                                                             | Reviewer:          | Size:<br><b>A4</b>                  | Sheet:<br><b>20</b> of <b>23</b> |

|                                                                          |                                                           |                               |                                                                                                                                                    |                                         |                              |                                 |
|--------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------|---------------------------------|
| Comments:                                                                | Company:<br><div>Willas</div>                             |                               | <div><div> 威倫電子股份有限公司<br/>Willas Electronic Corp.</div></div> |                                         | Variant:<br><div>DRAFT</div> | Git Hash:<br><div>cc04374</div> |
|                                                                          | Board Name:<br><div>Motor Control Board</div>             |                               |                                                                                                                                                    | Project Name:<br><div>Master Hand</div> |                              |                                 |
| Sheet Title:                                                             | File Name:<br><div>Misc - Holes Fiducials.kicad_sch</div> | Designer:<br><div>Damon</div> | Date:<br><div>2026-06-27</div>                                                                                                                     | Revision:<br><div>+ (Unreleased)</div>  |                              |                                 |
| Sheet Path:<br><div>/Project Architecture/Misc - Holes, Fiducials/</div> |                                                           | Reviewer:                     | Size:<br><div>A4</div>                                                                                                                             | Sheet:<br><div>20 of 23</div>           |                              |                                 |

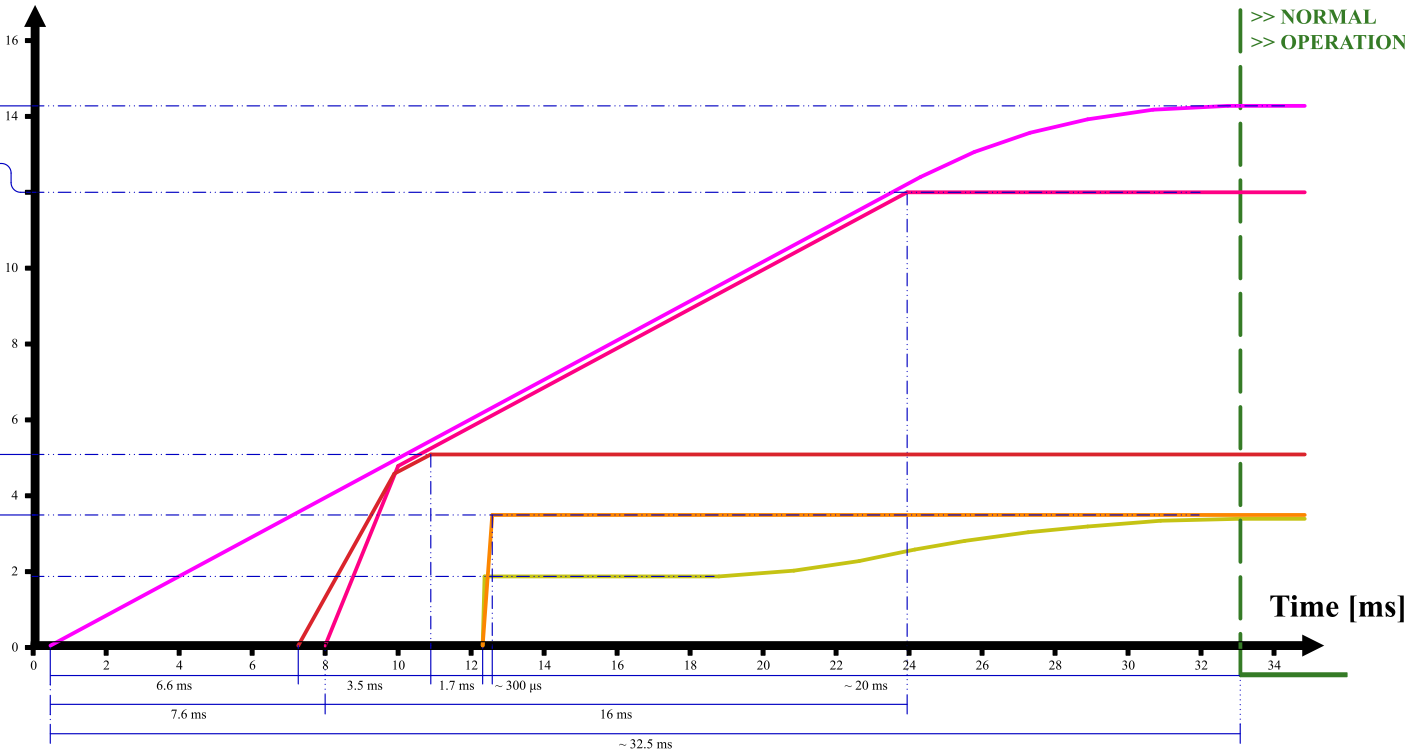
# [21] Misc - Board Version, Switch



|                                                                    |                                                                                                                                                       |                    |                     |                                     |                      |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------|-------------------------------------|----------------------|
| Comments:                                                          | Company:<br><div>Willas</div> <div>Willas Electronic Corp.</div> |                    | Variant:<br>DRAFT   |                                     | Git Hash:<br>cc04374 |
|                                                                    | Board Name:<br><b>Motor Control Board</b>                                                                                                             |                    |                     | Project Name:<br><b>Master Hand</b> |                      |
| Sheet Title:                                                       | File Name:<br>Misc - Board Version, Switch.kicad_sch                                                                                                  | Designer:<br>Damon | Date:<br>2026-06-27 | Revision:<br>+ (Unreleased)         |                      |
| Sheet Path:<br>/Project Architecture/Misc - Board Version, Switch/ |                                                                                                                                                       | Reviewer:          | Size:<br><b>A4</b>  | Sheet:<br><b>21</b> of <b>23</b>    |                      |

| NAME | SOURCE | LEVEL       |
|------|--------|-------------|
| +V?? | ??     | ? - ? ± ??% |
| +V?? | ??     | ? - ? ± ??% |
| +V?? | ??     | ? - ? ± ??% |
| +V?? | ??     | ? - ? ± ??% |
| +V?? | ??     | ? - ? ± ??% |

Voltage [V]



|                                     |                                            |  |                                                                                                                                                              |                              |                             |                      |
|-------------------------------------|--------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------|----------------------|
| Comments:                           | Company:<br>Willas                         |  | <div><div></div><div>威倫電子股份有限公司<br/>Willas Electronic Corp.</div></div> |                              | Variant:<br>DRAFT           | Git Hash:<br>cc04374 |
|                                     | Board Name:<br>Motor Control Board         |  |                                                                                                                                                              | Project Name:<br>Master Hand |                             |                      |
| Sheet Title:                        | File Name:<br>Power - Sequencing.kicad_sch |  | Designer:<br>Damon                                                                                                                                           | Date:<br>2026-06-27          | Revision:<br>+ (Unreleased) |                      |
| Sheet Path:<br>/Power - Sequencing/ |                                            |  | Reviewer:                                                                                                                                                    | Size:<br>A4                  | Sheet:<br>22 of 23          |                      |

|                                   |                                          |                    |                                                                                                                             |                              |                      |
|-----------------------------------|------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|
| Comments:                         | Company:<br>Willas                       |                    |  威倫電子股份有限公司<br>Willas Electronic Corp. | Variant:<br>DRAFT            | Git Hash:<br>cc04374 |
|                                   | Board Name:<br>Motor Control Board       |                    |                                                                                                                             | Project Name:<br>Master Hand |                      |
| Sheet Title:                      | File Name:<br>Revision History.kicad_sch | Designer:<br>Damon | Date:<br>2026-06-27                                                                                                         | Revision:<br>+ (Unreleased)  |                      |
| Sheet Path:<br>/Revision History/ |                                          | Reviewer:          | Size:<br>A4                                                                                                                 | Sheet:<br>23 of 23           |                      |